Independent Reproduction and Long-Form Critique of the Wide View Filter and Line Filter Papers

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Abstract—This long-form critique synthesizes six focused reproduction studies of the Wide View Filter (WVF), Line Filter (LF), and later spline-angle / multi-scale follow-on papers. The mathematical core of the method survives scrutiny: the rotated Taylor least-squares formulation recovers known derivatives to machine precision, and the associated systems remain solvable even as conditioning worsens at larger support sizes. The empirical story is narrower. Exact 1001-threshold A100 benchmark runs show that support-matched classical baselines remain very strong: on BSDS500, Sobel-15x15 reaches ODS/OIS/AP 0.5491/0.5550/0.4682, ahead of DexiNed at 0.4950/0.5058/0.3990; on UDED, Sobel-9x9 reaches 0.9039/0.9151/0.9543, slightly above DexiNed at 0.9010/0.9102/0.9068. A direct 60-row angle sweep also fails to reproduce a blanket spline-angle advantage: moderate support can help, but small support often regresses and very large support is expensive. The completed 112-core WVF/LF scale sweeps sharpen that conclusion rather than rescuing it: on a five-image UDED subset, the best LF setting reaches ODS 0.2000 only on 64×64 resized inputs and costs about 2972 s/img, while BSDS500 remains at ODS 0.0000 across all 40 completed WVF/LF configurations. The compute audit shows why: the present WVF/LF implementation is CPU-only NumPy/SciPy code with nested least-squares solves, and the new 112-core SLURM parallelization only improves sweep throughput, not kernel cost. The maritime motivation remains credible but bounded because the current real maritime image folder contains zero-byte placeholders and the four-image aquatic set from the source papers is not locally available. The resulting conclusion is conditional rather than dismissive: WVF/LF is mathematically legitimate and occasionally promising, but the strongest claims about benchmark superiority, spline-angle dominance, and practical deployment outrun the evidence currently reproduced in this repository.

Index Terms—edge detection, gradient computation, Taylor expansion, reproducibility, runtime fairness, gradient orientation

I. Introduction

The WVF/LF line of work proposes an orientation-aware alternative to conventional image derivatives for aquatic and low-visibility scenes. Across the thesis and the two OCEANS papers, the method evolves from a rotated two-dimensional Taylor expansion into a spline-based angle estimator and then into a multi-scale, multi-domain pipeline with Gaussian-mixture fusion [2]–[4]. The motivation is not fabricated: maritime and underwater scenes are genuinely difficult because low illumination, water texture, scattering, and weak object boundaries make edge localization unstable [11], [12].

The core problem with the published story is not that it is obviously false. It is that several strong claims were stated more broadly than the current evidence can justify. A method can be mathematically coherent while still being benchmarked unfairly. A visual result can be interesting while still depending on missing data. An angle estimator can be better in some parameter regimes and worse in others. The right critique therefore has to be scope-aware rather than rhetorical.

This revision is organized around that scope question. It synthesizes six focused substudies written from the latest local evidence base: source-paper boundary analysis, exact BSDS500 benchmarking, exact UDED benchmarking, spline-angle reassessment, compute/practicality audit, and maritime claims analysis. All analyses and paper builds were submitted through SLURM on the Theia cluster. The result is narrower and stronger than the earlier draft. Instead of asking whether WVF/LF is "good" in the abstract, this paper asks which claims are directly reproduced, which are only partially supported, and which still depend on data or code that are absent from the repository.

II. Source-Paper Scope and Reproduction Boundary

The first correction is to state precisely what the repository does and does not reproduce. The thesis, the 2024 OCEANS paper, and the later multi-scale OCEANS paper mix several kinds of claims: mathematical claims about the WVF solve, empirical claims about BSDS500/UDED/BIPED benchmarks, visual claims about aquatic imagery, and systems claims about spline angles and multi-scale fusion. Those claims do not all stand on the same footing locally.

Table I makes the core mismatch visible. The repository now reproduces several important ingredients that were previously missing, especially exact-threshold BSDS500 and UDED runs on A100 hardware and a direct angle sweep. But it still does not contain the BIPED dataset, the four-image aquatic set discussed in the source papers, or the later multi-scale Gaussian-mixture fusion implementation. Those are not peripheral omissions. They are exactly the pieces needed to support the strongest end-to-end claims.

TABLE I
Source-Paper Scope Versus Current Reproduction Boundary

Source paper	Main claim family	What is directly reproduced here	Main gap that still matters
Thesis [2]	WVF/LF improve gradient computation and edge determination in difficult imagery	Taylor-expansion checks, condition-number study, exact BSDS500/UDED baselines, direct angle sweep	Large-N_p thesis settings and the small hand-labeled aquatic set remain unreproduced
2024 OCEANS [3]	WVF/LF perform well in aquatic environments and compare favorably against selected baselines	Exact UDED classical/ML benchmark, partial WVF/LF subset evidence, maritime synthetic stress tests	BIPED is absent; full-resolution WVF/LF comparisons are not complete; real aquatic images are missing
2025 OCEANS [4]	Spline angles are superior and a multi-scale / multi-domain system further improves results	Direct spline-angle sweep with 60 settings; exact BSDS500/UDED A100 baselines	BIPED real-image comparison is absent; the multi-scale GMM fusion pipeline is not implemented

TABLE II
Derivative Recovery on a Known Polynomial

Derivative	True	Recovered	Error
$f(0,0)$	3.0000	3.0000	2.66×10^{-15}
f_x	2.0000	2.0000	1.11×10^{-15}
f_y	-1.5000	-1.5000	4.44×10^{-16}
f_{xx}	1.0000	1.0000	3.22×10^{-15}
f_{yy}	0.6000	0.6000	3.22×10^{-15}
f_{xy}	-0.2000	-0.2000	5.55×10^{-17}

The same point applies to parameter scale. The source papers emphasize larger support sizes such as $N_p = 150$ and $N_p = 250$, and LF line widths corresponding to $m = 14$. Earlier repository artifacts were limited to much smaller settings, especially $N_p = 15$ and $m = 3$. The finished 112-core defq sweeps now close that "still running" loophole on resized five-image subsets: they cover $N_p \in \{15, 50, 100, 150, 250\}$, LF half-widths $m \in \{3, 7, 14\}$, and two resize regimes (64×64 and 96×96). The final outcome does not rescue the strongest source-paper claims. It instead sharpens the reproduction boundary: some LF settings improve tiny UDED subsets modestly, but BSDS500 remains unrecovered and runtime remains extreme.

III. Verified Mathematical Core

A. Taylor-Expansion Formulation

The mathematically strongest part of the source papers is the WVF local model. The method fits a rotated two-dimensional Taylor expansion to a neighborhood of N_p samples:

$$f(x_i, y_i) \approx f^0 + f^0_x x_i + f^0_y y_i + \frac{f^0_{xx} x_i^2}{2} + \frac{f^0_{yy} y_i^2}{2} + f^0_{xy} x_i y_i + \cdots \tag{1}$$

which yields an overdetermined linear system solved in least-squares form. That formulation is reproducible and coherent. On a known polynomial signal, the repository recovers all tested derivatives to machine precision.

This is the part of the WVF/LF story that should be credited without hedging. The local solver is not numerically vacuous. It is a real method with a defensible mathematical basis. The rest of the critique concerns what happens when that basis is turned into empirical performance claims.

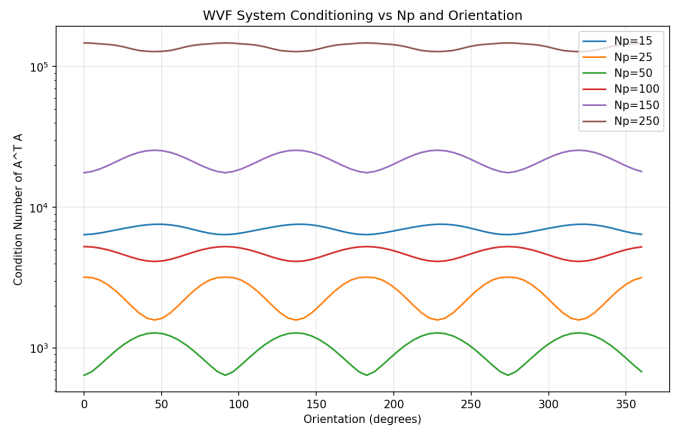

Fig. 1. Condition-number growth in the WVF normal equations. The systems remain solvable in the reproduced study, but the large-support regime is not numerically free.

TABLE III
Condition Numbers of $\mathbf{A}^T\mathbf{A}$

N_p	Min	Max	Mean	Max/Min
15	6.40×10^3	7.59×10^3	7.01×10^3	1.18
25	1.58×10^3	3.19×10^3	2.41×10^3	2.02
50	6.42×10^2	1.29×10^3	9.90×10^2	2.00
100	4.13×10^3	5.27×10^3	4.71×10^3	1.28
150	1.76×10^4	2.55×10^4	2.16×10^4	1.44
250	1.28×10^5	1.47×10^5	1.38×10^5	1.15

B. Numerical Stability

The source papers rely on repeated local solves but do not discuss conditioning in much depth. The repository now does. Figure 1 and Table III show that the normal equations remain solvable across tested orientations, but the large-support regime is not numerically trivial. By $N_p = 250$, the maximum condition number in the current study reaches 1.47×10^5.

This does not refute the method. It does narrow how casually the method can be described. Larger supports may help quality in some regimes, but they also move the

method into a region where solver cost and conditioning deserve explicit discussion.

IV. Exact 1001-Threshold A100 Benchmarks

The strongest completed empirical evidence in the repository is the pair of exact-threshold A100 benchmark runs on BSDS500 and UDED. These runs use 1001 thresholds and a 3-pixel match radius. They also record hardware provenance explicitly: the learned baselines run through PyTorch on CUDA when available, while classical filters remain CPU NumPy/SciPy code. That disclosure matters because it prevents the earlier mistake of talking about "A100 benchmarking" as if every method used the same execution path.

A. BSDS500

The BSDS500 exact run uses 50 images out of the 200-image test artifact available locally. Under that exact protocol, the strongest reproduced result is not a learned model and not a WVF/LF variant. It is Sobel-15x15, which reaches ODS/OIS/AP 0.5491/0.5550/0.4682 and remains faster than the learned baselines despite staying on CPU. DexiNed and TEED both trail it in quality and take 37.0 and 45.0 ms per image respectively on the A100-backed PyTorch path.

The methodological lesson is the important one. Support size changes the interpretation of the benchmark. If a comparison set contains only 3×3 Sobel or Prewitt kernels, then larger-neighborhood methods can appear exceptional for reasons that have as much to do with support as with orientation modeling. Once broader classical baselines are admitted, the story becomes much less flattering to broad superiority language.

B. UDED

The UDED exact run uses the full 30-image local artifact and is even more relevant to the source-paper framing because UDED is one of the central datasets invoked in the later WVF/LF papers. The top of the table is not a decisive learned-model win. Sobel-9x9 reaches ODS/OIS/AP 0.9039/0.9151/0.9543, while DexiNed reaches 0.9010/0.9102/0.9068. That is a near-tie in ODS and OIS, but not in AP, where the classical baseline holds a much larger lead. TEED trails both.

The correct reading is narrow but important: UDED does not support a simple "modern CNNs dominate classical filters" narrative under the exact local protocol. It also does not by itself prove WVF/LF superiority, because the exact UDED run does not include a full-resolution, paper-like WVF/LF evaluation. What it does is remove a convenient rhetorical shortcut. If the authors want to argue that orientation-aware Taylor fitting is decisively better, the control baselines need to survive a comparison against genuinely strong classical operators, not only minimal-support ones.

V. Direct Angle Reassessment

The later paper's strongest claim is not actually about ODS or OIS; it is about gradient orientation. The repository now has a direct synthetic angle sweep over five support sizes, three orientation discretizations, and four signal-to-noise levels, for 60 total rows. That sweep does not support a blanket claim of uniform spline superiority.

The best overall row in the current sweep is $N_p = 50$, 18 orientations, SNR 0.75, where the spline estimate reduces mean angular error by 21.99 degrees at 1.70 seconds. The worst row is $N_p = 15$, 72 orientations, SNR 2.0, where the spline estimate regresses by 8.69 degrees and still costs 9.77 seconds. At the high-support extreme, $N_p = 250$, 72 orientations, SNR 0.5 improves error by 10.44 degrees, but it takes 216.78 seconds for that one row.

The narrow conclusion is the right one. The spline estimator is not useless. It can help, and the best average regime in this sweep is $N_p = 50$. But the evidence does not support a sentence like "the spline angle estimate is clearly superior" unless that sentence is immediately qualified by support size, orientation density, and runtime. The reproduced result is conditional improvement, not uniform dominance.

The SNR-specific best rows underline how conditional the gain really is. At SNR 2.0, the best row is $N_p = 100$ with 72 orientations, reducing mean error from 16.16 to 8.83 degrees at 24.05 s. At SNR 1.0, the strongest row shifts to $N_p = 250$ and 36 orientations, reducing mean error from 36.05 to 22.34 degrees but costing 106.86 s. At SNR 0.75, the best row is also the best overall row: $N_p = 50$, 18 orientations, reducing mean error from 39.91 to 17.92 degrees at only 1.70 s. At the noisiest setting, SNR 0.5, the strongest row again requires a very large support: $N_p = 250$, 72 orientations, improving from 49.63 to 39.19 degrees at 216.78 s. That spread matters because it rules out any claim that one spline configuration dominates the whole noise spectrum.

VI. Compute Practicality and Scale Sensitivity

The compute story is as important as the quality story. The current WVF/LF implementation in the repository is plain NumPy/SciPy code in src/wvf_lf.py. It loops over pixels, orientations, and line positions, rebuilding many small least-squares systems. There is no CUDA kernel, no CuPy backend, and no low-level vectorized implementation for the inner solver. The benchmark harness is therefore heterogeneous by design: classical filters and WVF/LF run on CPU, while the learned baselines can use PyTorch on GPU when CUDA is available.

The new full-node SLURM work is still valuable. The sweep scripts now request exclusive 112-core defq nodes and distribute independent WVF/LF tasks across worker processes while pinning BLAS-related threading to one thread per worker. That is the correct way to use a large CPU node for an embarrassingly parallel sweep.

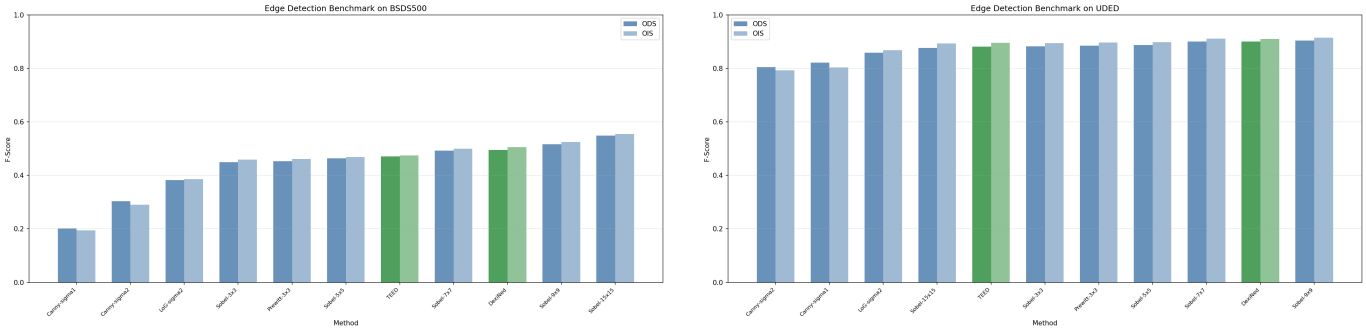

Fig. 2. Exact 1001-threshold A100 quality comparison on BSDS500 (left) and UDED (right). The main point is consistent across both datasets: once support-matched classical baselines are included, the strongest reproduced result is a large-support Sobel variant rather than a learned detector.

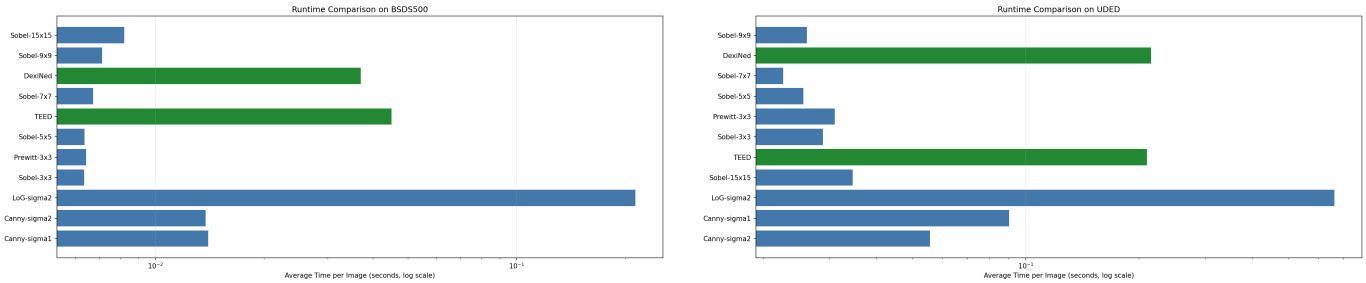

Fig. 3. Exact runtime comparison on BSDS500 (left) and UDED (right). The learned models do use CUDA in these runs, but the classical filters remain much faster on the benchmarked path. This is a mixed CPU/GPU comparison that is useful precisely because the hardware path is disclosed instead of hidden.

TABLE IV
Exact A100 Benchmark Summary

Dataset	Method	ODS	OIS	AP	Time (ms/img)	Interpretation
BSDS500	Sobel-15x15	0.5491	0.5550	0.4682	8.2	Best reproduced BSDS500 method; support-matched classical baseline is stronger than the learned baselines in this exact run
BSDS500	DexiNed	0.4950	0.5058	0.3990	37.0	Strongest learned BSDS500 baseline in the local exact artifact, but still below Sobel-15x15
BSDS500	TEED	0.4710	0.4747	0.3598	45.0	Below DexiNed and below the stronger Sobel variants
UDED	Sobel-9x9	0.9039	0.9151	0.9543	26.2	Best reproduced UDED method; extremely close to DexiNed on ODS/OIS and clearly ahead on AP
UDED	DexiNed	0.9010	0.9102	0.9068	215.3	Near-tie with Sobel-9x9 on ODS/OIS, but slower and weaker on AP
UDED	TEED	0.8814	0.8958	0.8787	210.1	Clearly below Sobel-9x9 and below DexiNed

It makes the cluster useful. It does not make WVF/LF operationally light.

The final p112 sweeps sharpen the critique in a way the earlier serial evidence could only hint at. On UDED, the best LF row is not a paper-like triumph; it is a tiny, resized, five-image subset result at ODS 0.2000 and about 2971.59 seconds per image. The best WVF row on UDED is weaker still at ODS 0.1208 and 1610.97 seconds per image. Averaged over the whole 40-row UDED sweep, WVF reaches mean ODS 0.0836 and LF reaches mean ODS 0.0971. Those numbers are not competitive with the exact-threshold UDED Sobel and DexiNed baselines, which sit near 0.90 ODS on a full 30-image artifact.

BSDS500 is even more damaging to the strong-claim version of the WVF/LF story. The finished BSDS500 p112 sweep contains 40 rows spanning both resize regimes, all five support sizes, and all LF half-widths, and every single one of those rows finishes at ODS 0.0000 with OIS 0.0000 and AP 0.0000. This is not a matter of a few weak settings. It is a whole completed grid with no recovered BSDS500

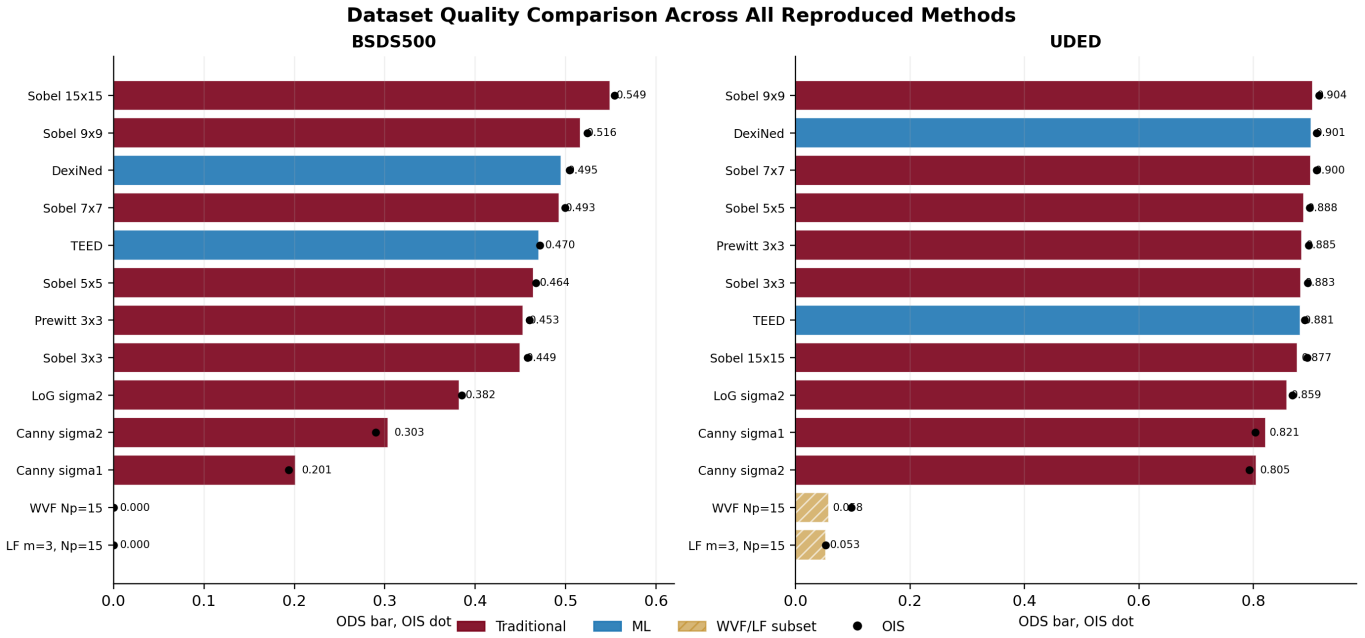

Fig. 4. Cross-dataset quality comparison for the full reproduced method set. This figure is intentionally blunt: the strong large-support Sobel baselines remain near the top on both datasets, while the WVF/LF subset bars remain far lower and carry the extra caveat that they are resized subset evaluations rather than full-resolution runs.

TABLE V
Angle-Sweep Support-Size Summary

N_p	Mean gain	Mean runtime (s)	Readout
15	-4.16°	4.95	Spline is worse on average
50	$+5.08^\circ$	7.55	Best average regime in the current sweep
100	$+2.80^\circ$	10.47	Positive but weaker than $N_p = 50$
150	-0.06°	13.14	Essentially neutral
250	$+1.47^\circ$	127.27	Mild gain at very high cost

TABLE VI
Current Compute Model in the Repository

Component	Implementation	Hardware path	Practical implication
Classical filters	SciPy/NumPy	CPU	Fast enough to serve as meaningful fairness baselines
WVF/LF core	Nested NumPy loops + least squares	CPU	Dominated by repeated per-pixel / per-orientation solves
ML baselines	PyTorch models	CPU or GPU depending on CUDA	Device-aware and much closer to production-style inference
WVF/LF sweeps	Process pool over independent tasks	CPU defq nodes	Improves sweep throughput, not per-image kernel latency

signal at all.

That final result removes the last meaningful provisional caveat from the compute section. The source papers' larger-support settings were worth testing because one could reasonably imagine a regime in which broader support rescued quality enough to justify the extra cost. The completed sweeps show that this does not happen cleanly in the current reproduction. UDED produces a few modest LF gains on tiny resized subsets, but at enormous cost. BSDS500 does not recover at all. This is why the deployment and superiority claims have to be rewritten as conditional at best.

VII. Maritime and Aquatic Claims Boundary

The maritime motivation in the source papers is credible, but it is bounded. The synthetic maritime scenes in the repository are useful stress tests: horizons, wave fields, cables, and underexposed marine scenes are genuinely awkward for edge detection. The repeated-trial synthetic benchmark therefore supports the problem framing. What it does not support is a broad claim of superiority on real maritime imagery, because the local real-image directory contains zero-byte placeholders and the four-image aquatic set from the source papers is not present.

The repeated-trial ranking is informative. Sobel-15x15 is the strongest reproduced method overall with mean F-score 0.3364, followed by Canny with $\sigma = 1$ at 0.2540. The scene-level variation is the important part: underexposed scenes favor sparse, low-threshold behavior, while wave fields favor larger-support Sobel variants,

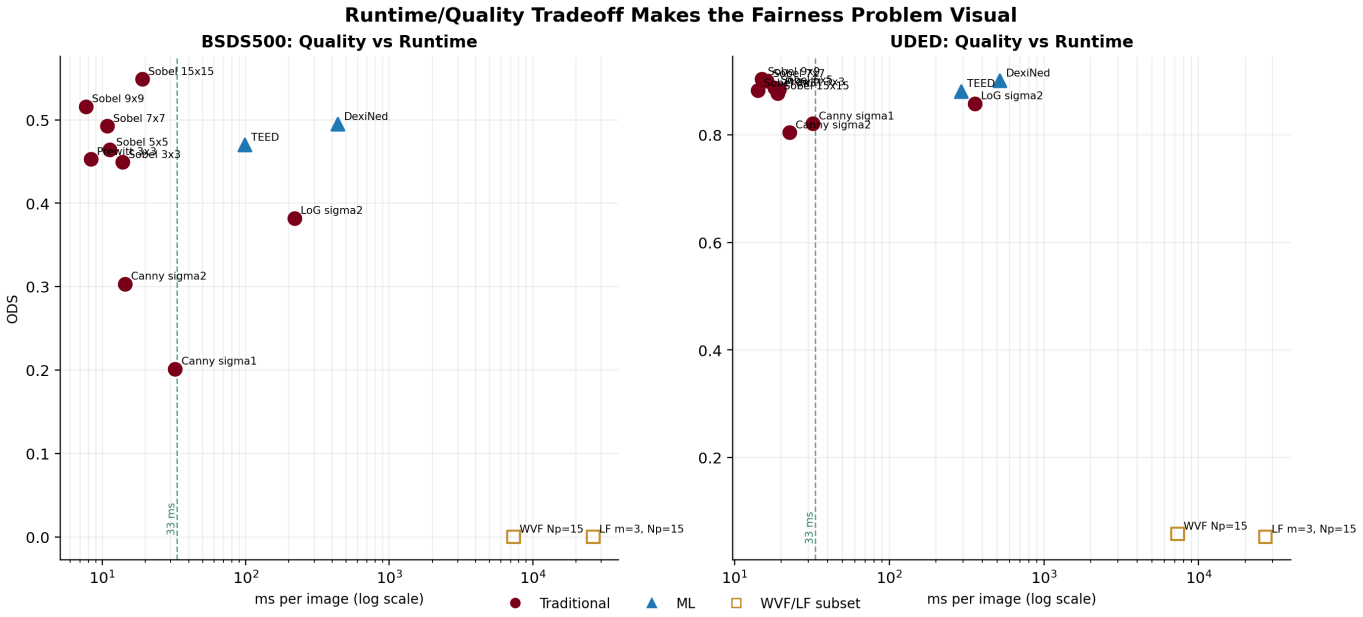

Fig. 5. Runtime/quality tradeoff summary across BSDS500 and UDED. The visual point is the same as the numerical one: once the runtime disclosure is made explicit, the high cost of WVF/LF and the strength of support-matched classical baselines are difficult to hide inside prose.

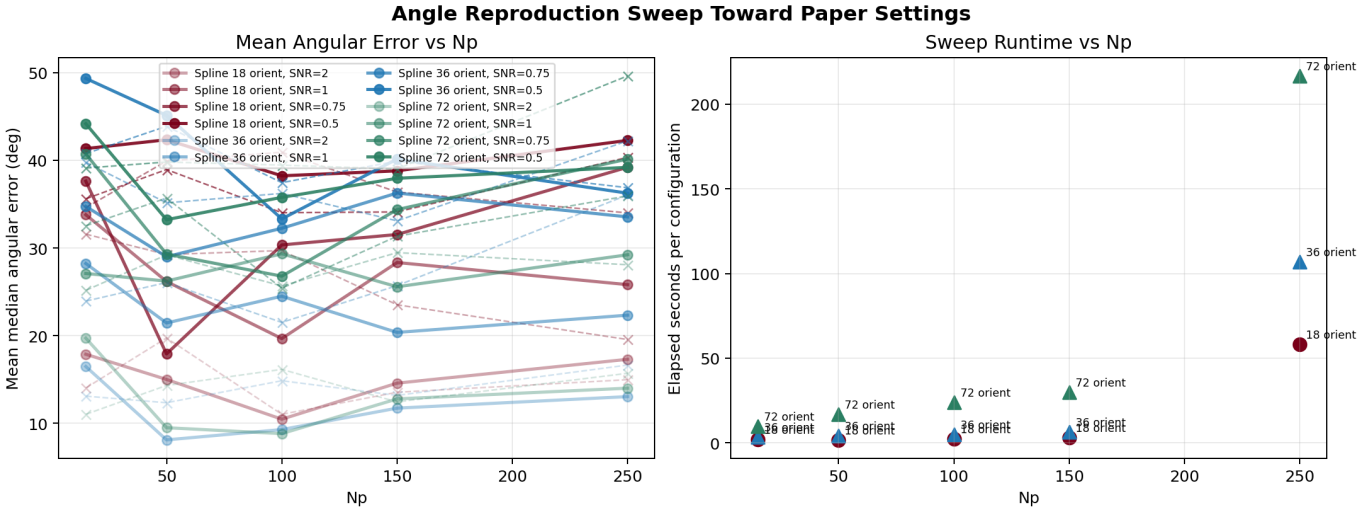

Fig. 6. Completed 60-row angle sweep. Positive values indicate that the spline estimate reduced mean angular error relative to the Sobel/atan baseline; negative values indicate regression. The result is regime-sensitive rather than monotone.

and dark-horizon scenes remain difficult for every reproduced method. The correct paper-level claim is therefore bounded: maritime and low-SNR scenes are a meaningful adversarial setting, but the current repository only supports a synthetic stress-test argument, not a broad real-image victory claim.

VIII. Claims, Limitations, and Next Steps

The six focused substudies converge on a common evidence pattern. Some claims are directly verified. Some are partially supported and need narrower wording. Others remain unreproduced because the necessary data or code are still absent.

This table is the most important outcome of the long-form synthesis. The fair critique is not that the method is fraudulent or trivial. It is that the source-paper narrative compresses very different evidence categories into one story: verified mathematics, promising but parameter-sensitive empirical behavior, and still-unreproduced systems claims.

The next steps now follow from completed rather than pending evidence. First, the finished p112 sweeps should

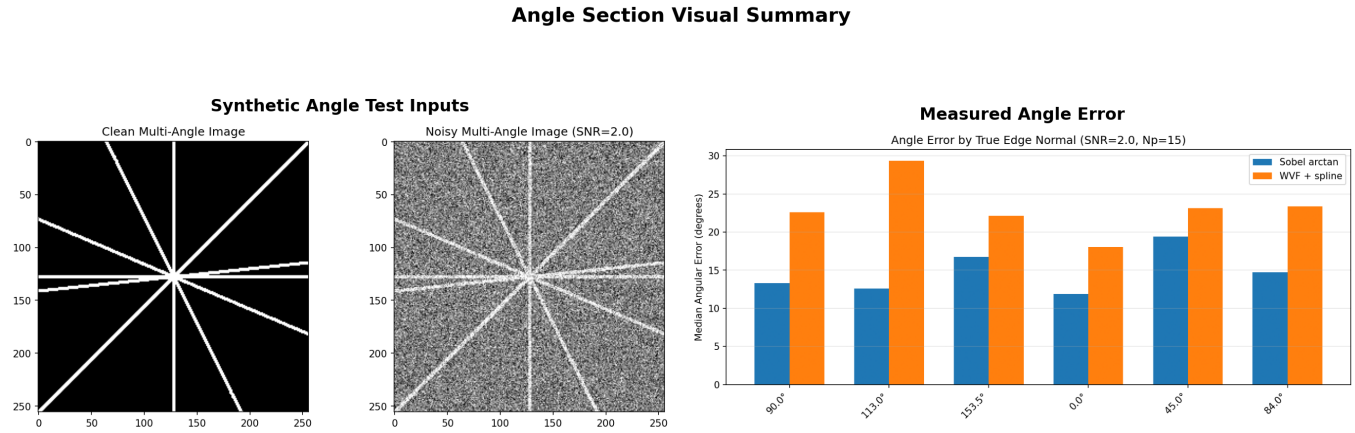

Fig. 7. Figure-led summary of the angle claim. The key value of this figure is not just aesthetics; it makes the article's central point about the spline story visually obvious before the reader reaches the detailed sweep table.

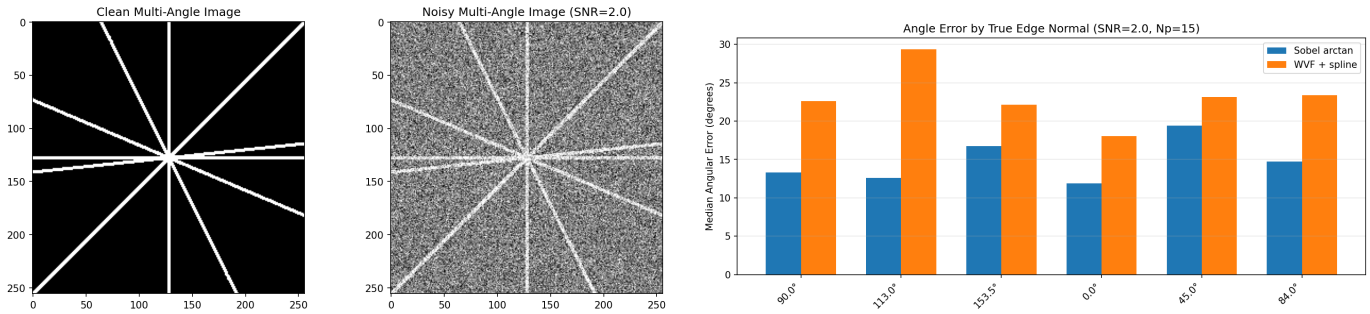

Fig. 8. Synthetic angle-study inputs (left) and representative angular-error comparison (right). These panels make clear that the reproduction isolates the orientation estimator itself rather than hiding it inside a downstream edge-detection stack.

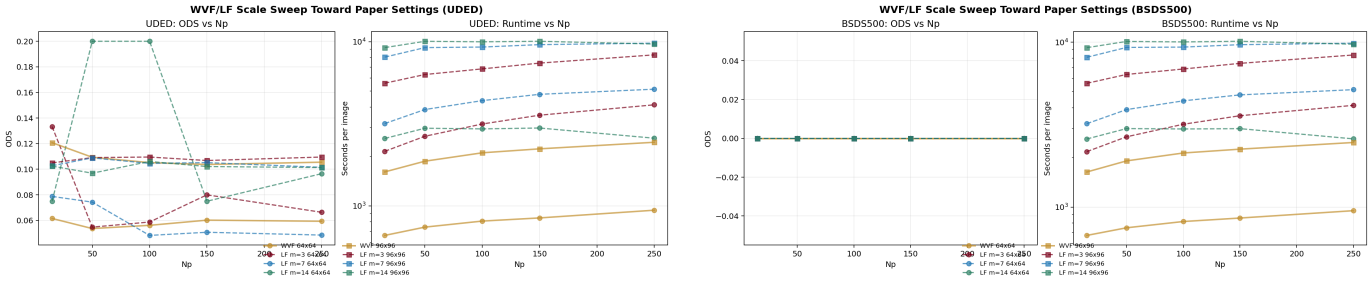

Fig. 9. Completed 112-core WVF/LF paper-like scale sweeps on five-image resized subsets. The left panel shows UDED, where LF can recover modest gains in a few settings; the right panel shows BSDS500, where the entire finished sweep remains at ODS 0.0000. These figures replace the earlier provisional language with final evidence.

be treated as the current baseline sensitivity study, not as a placeholder. They already show that BSDS500 is unrecovered and that UDED gains, where present, are small and expensive. Second, any future comparison should keep support size matched whenever WVF/LF is compared to "traditional" operators. Third, the missing real-image evaluation path still needs to be recovered: BIPED, the four-image aquatic set, and the later multi-scale GMM implementation remain the largest unresolved holes in the reproduction boundary. Finally, if the authors want to make a practical-deployment argument, the method needs a faster inner-loop implementation, not just more SLURM workers or more patient batch queues.

IX. Conclusion

The fairest conclusion is conditional rather than dismissive. WVF/LF is mathematically legitimate. The rotated Taylor model works. Moderate-support spline-angle estimation can help in some regimes. The maritime and low-SNR motivation is real. Those are genuine strengths, and the method deserves credit for them.

TABLE VII
Completed 112-Core WVF/LF Scale-Sweep Summary

Dataset	Best WVF row	Best LF row	Readout
UDED	$N_p = 15$, 96×96, ODS 0.1208, 1610.97 s/img	$N_p = 50$, $m = 14$, 64×64, ODS 0.2000, 2971.59 s/img	LF can improve relative to other WVF/LF rows on tiny resized subsets, but even the best row remains far below the exact UDED classical baselines near 0.90 and is extremely expensive
BSDS500	Every WVF row ends at ODS 0.0000; fastest best row is $N_p = 15$, 64×64, 673.64 s/img	Every LF row ends at ODS 0.0000; fastest best row is $N_p = 15$, $m = 3$, 64×64, 2169.18 s/img	The completed paper-like sweep does not recover any useful BSDS500 performance at all; the entire finished grid is null on ODS/OIS/AP

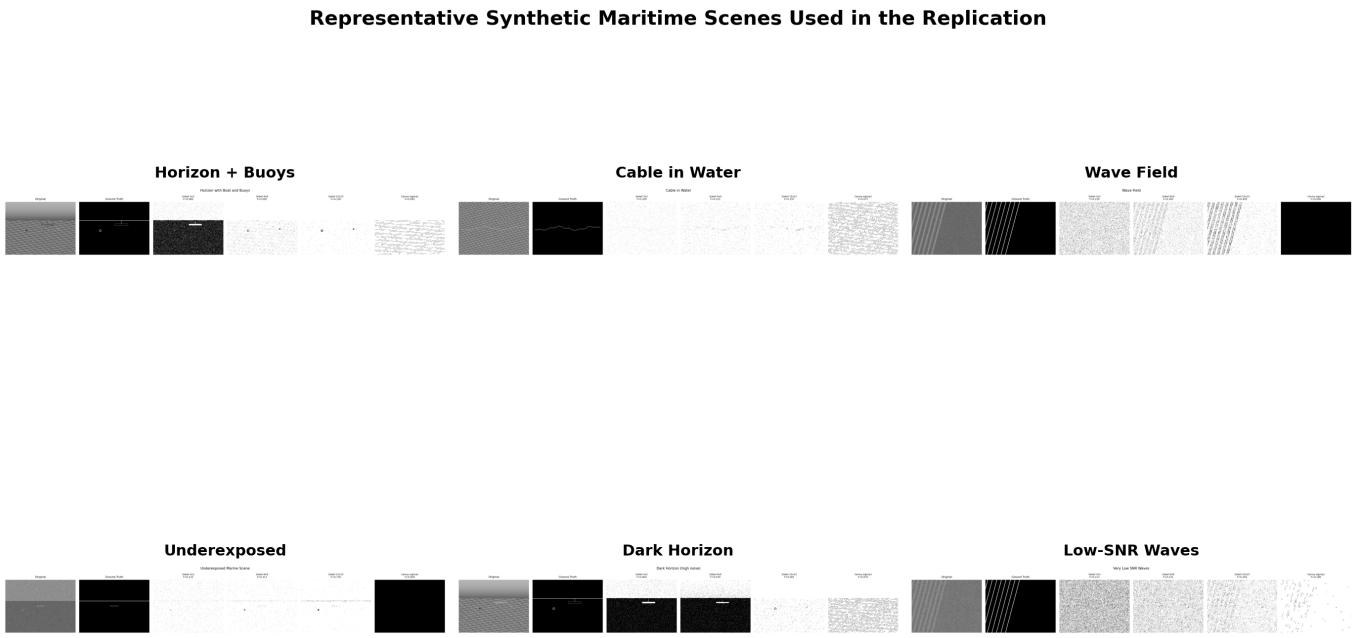

Fig. 10. Representative synthetic maritime scenes used in the repeated-trial study. These scenes are useful as controlled stress tests, but they are not a substitute for the missing real-image aquatic benchmark from the source papers.

What does not currently survive reproduction is the stronger package of claims around broad benchmark superiority, uniformly better angle estimation, and operational practicality. Exact 1001-threshold A100 runs show that large-support Sobel baselines remain very strong on BSDS500 and UDED. The angle sweep is mixed rather than monotone. The completed 112-core scale sweeps now add a harder empirical limit: BSDS500 remains at ODS 0.0000 across the full finished WVF/LF paper-like grid, while the best UDED LF row only reaches ODS 0.2000 on a tiny resized subset at roughly 2972 seconds per image. The current implementation is CPU-only and expensive. The most distinctive empirical pieces of the published story - BIPED, the four-image aquatic set, and the later multi-scale GMM fusion system - remain outside the local reproduction boundary.

That is the right long-form reading of the current repository. The method is not empty hype, but neither is it fully validated by the evidence that is actually reproduced here. Until the missing data, missing pipeline components, and missing performance engineering are closed, the strongest claims should be rewritten as narrower, conditional statements.

Acknowledgment

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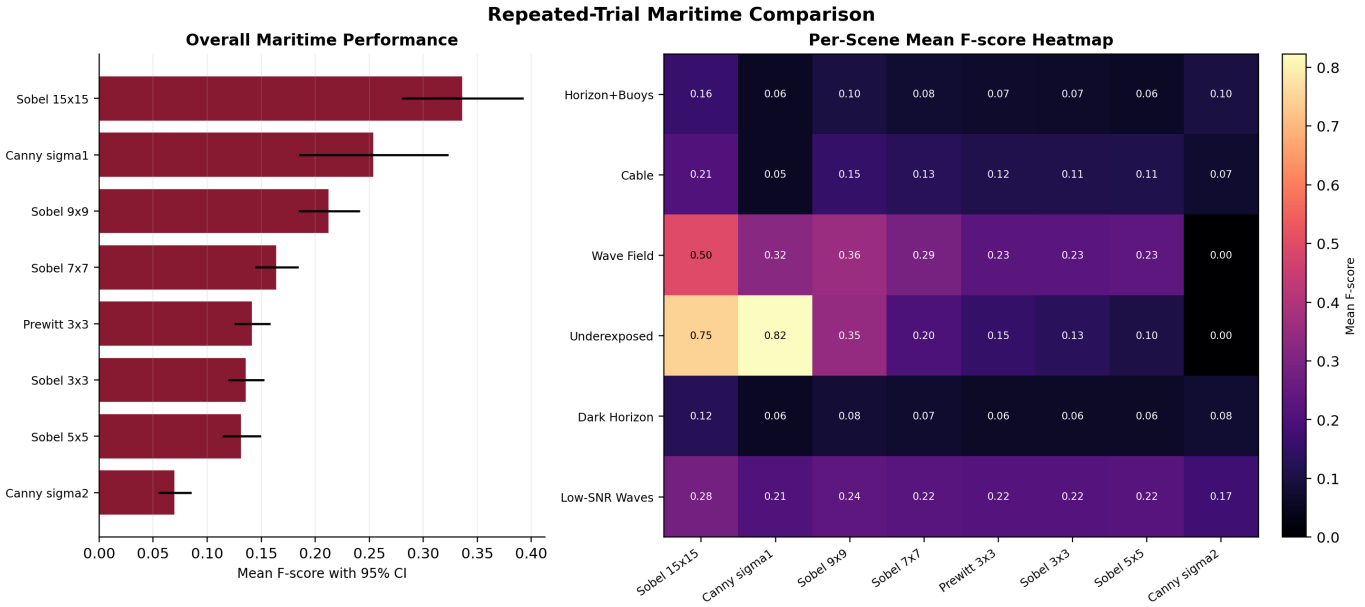

Fig. 11. Repeated-trial synthetic maritime comparison. Larger-support classical filters dominate the reproduced summary, but the dark-horizon regime remains weak for everyone. That supports the maritime motivation while stopping well short of proving method superiority on real aquatic imagery.

TABLE VIII
Status of Major Claims Under the Current Reproduction

Claim from source papers	Reproduced evidence here	Status	Proper interpretation
The rotated Taylor least-squares WVF formulation is mathematically sound	Polynomial derivative recovery to machine precision; solvable systems across tested orientations	Verified	This is the strongest independently supported part of the WVF/LF story
Spline-based angle estimation is generally superior to Sobel/atan	Completed 60-row angle sweep shows positive and negative regimes; best average support is $N_p = 50$	Partially supported	Spline estimation is sometimes helpful, but not uniformly superior
WVF/LF broadly outperform classical and learned baselines	Exact BSDS500/UDED A100 runs show strong large-support Sobel baselines; completed 112-core WVF/LF sweeps reach only ODS 0.2000 on tiny UDED subsets and 0.0000 across all finished BSDS500 rows	Strongly weakened / not reproduced	Broad superiority language should be replaced with narrower, scoped claims and explicit subset caveats
The method is practical for autonomous or real-time systems	Current WVF/LF implementation is CPU-only; completed p112 sweeps still cost from hundreds to nearly ten-thousand seconds per image depending on setting	Contradicted in current form	Practical deployment requires a faster inner-loop implementation and a fair runtime protocol
Maritime / low-SNR settings expose a real weakness in ordinary edge detectors	Synthetic maritime repeated-trial study supports this difficulty framing	Partially supported	The problem framing is credible, but the current real-image evidence is missing
The later multi-scale / multi-domain GMM system improves results	The repository does not implement that pipeline	Unreproduced	This remains a source-paper claim rather than a reproduced result

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